CYBERSECURITY PORTFOLIO PROJECT

SOC Analyst Home Lab
Master Project Summary

Building a Splunk SIEM for Windows Security Monitoring, Detection & Investigation

Norbert Endoh

Project Role: SOC Analyst / Lab Owner
Personal Controlled Cybersecurity Home Lab

Completed October 2026

Portfolio purpose	Hands-on SOC Analyst evidence for resume, LinkedIn, portfolio/GitHub, and interviews
Platform	VMware Workstation
SIEM	Splunk Enterprise 10.4.3 on Ubuntu Server
Monitored endpoint	Windows 10 — DESKTOP-4AMTH10
Analyst workstation	Kali Linux / Firefox
Log collection	Splunk Universal Forwarder → TCP 9997 → Splunk Enterprise
Network	Bridged
Project status	Complete for the defined scope

1. Project Purpose

This project was built to demonstrate an end-to-end SOC Analyst workflow in a controlled home lab. The objective was not simply to install Splunk. The project established centralized Windows security monitoring, validated telemetry ingestion, analyzed Windows Security events with SPL, engineered detections, configured scheduled alerts, built an analyst dashboard, investigated a triggered alert, correlated authentication and process activity, documented the incident disposition, and performed a final operational health check.

2. Lab Architecture

- Windows 11 physical host running VMware Workstation.
- Windows 10 VM as the monitored endpoint.
- Ubuntu Server VM running Splunk Enterprise.
- Splunk Universal Forwarder on Windows 10 forwarding telemetry to Splunk over TCP 9997.
- Kali Linux VM used as the SOC analyst workstation to access Splunk Web.
- Metasploitable 2 and Kioptrix Level 1 retained as additional controlled lab targets.
- VMs intentionally configured with Bridged networking.

3. End-to-End SOC Workflow Completed

- Designed the lab architecture and established the Splunk SIEM server.
- Configured Windows Security log collection and validated endpoint-to-SIEM ingestion.
- Analyzed Windows Security events using Splunk Search Processing Language (SPL).
- Developed Detection Use Case 01 for repeated failed Windows logons using Event ID 4625.
- Refined the detection to handle multivalue account fields and generalized it beyond the test account.
- Operationalized the rule as a scheduled alert and added per-user suppression to reduce duplicate noise.
- Built the SOC Windows Security Monitoring Dashboard for authentication, account-management, privileged-logon, process, and PowerShell visibility.
- Validated five enabled security alerts.
- Generated controlled failed-logon activity and verified automatic alert triggering.
- Investigated three failed logons followed by a successful interactive logon for labuser.
- Reviewed Event ID 4688 process activity and performed a targeted PowerShell check.
- Closed the incident as True Positive — Authorized/Benign Security Test.
- Performed final health validation of telemetry ingestion, detections, and dashboard visibility.

4. Key Detection & Investigation Evidence

Primary detection	3 or more failed Windows logons for the same user within 5 minutes
Primary event	Windows Security Event ID 4625
Scheduled execution	Every 5 minutes
Operational search window	Last 15 minutes
Suppression	15 minutes by target_user
Alert severity at final validation	High
Investigated account	labuser
Source	127.0.0.1
Workstation	DESKTOP-4AMTH10
Authentication correlation	3 failed 4625 events followed by successful 4624 interactive logon

Process investigation	Windows Security Event ID 4688
Targeted PowerShell finding	No labuser PowerShell execution found in the investigated 30-minute window
Incident disposition	True Positive — Authorized/Benign Security Test

5. Final Operational Validation

- 114 Windows Security events were received during the final 15-minute ingestion validation window.
- Five Splunk security alerts were confirmed enabled.
- The SOC dashboard remained operational and displayed current authentication and process telemetry.
- The final dashboard view showed 6 failed Windows logons and 165 successful Windows logons in the selected Last 24 hours window.
- The project demonstrated the complete path: ingestion → detection → alerting → investigation → disposition → visualization.

6. Portfolio Document Set

Document	Title	What It Proves
1	Foundation, Build & Early SOC Analysis	Architecture, networking, Splunk deployment, Universal Forwarder, telemetry ingestion, and early event analysis.
2	Detection Engineering — Use Case 01	Baseline, controlled Event ID 4625 generation, SPL detection logic, scheduled alerting, validation, and suppression/tuning.
3	Dashboard, Incident Investigation & Final Validation	SOC dashboard, enabled alerts, triggered-alert investigation, authentication/process correlation, incident closure, and final health validation.

7. SOC Analyst Skills Demonstrated

- Splunk Enterprise SIEM and Splunk Universal Forwarder
- Windows Security Event Log monitoring and analysis
- SPL searching, statistics, filtering, and field normalization
- Event IDs 4624, 4625, 4672, 4688, 4720–4726, 4732, and 4733
- Detection engineering and threshold-based logic
- Scheduled alerts, cron scheduling, severity, throttling, and suppression
- SOC dashboard development
- Alert triage and validation
- Authentication-event correlation
- Process-creation investigation
- Incident disposition, closure, and documentation
- SIEM health validation and evidence-based reporting

8. Resume-Ready Project Entry

SOC Analyst Home Lab — Splunk SIEM for Windows Security Monitoring, Detection & Investigation

- Built a VMware-based SOC lab with Splunk Enterprise on Ubuntu Server, a Windows 10 monitored endpoint, Splunk Universal Forwarder, and Kali Linux as the analyst workstation.
- Collected and analyzed Windows Security telemetry and developed a multi-panel SOC dashboard covering authentication, privileged logons, account-management activity, process creation, and PowerShell visibility.
- Engineered and validated a repeated-failed-logon detection using Event ID 4625, scheduled alerting, field normalization, and per-user suppression.
- Investigated a triggered alert by correlating failed and successful authentication events with Event ID 4688 process telemetry, then documented the incident disposition and closure.
- Validated the completed SIEM environment by confirming active telemetry ingestion, enabled detections, and functioning dashboard visibility.

9. Interview Summary

"I built an end-to-end Splunk SOC home lab with a Windows 10 endpoint forwarding Security logs to Splunk Enterprise on Ubuntu Server. I validated the telemetry pipeline, created SPL detections and scheduled alerts, built a Windows security monitoring dashboard, and tested a repeated-failed-logon use case. When the alert fired, I investigated the failed logons, correlated them with a successful interactive login, reviewed Event ID 4688 process activity, and checked specifically for PowerShell execution. Because I intentionally generated the activity in my controlled lab, I documented it as a true-positive detection with an authorized/benign test disposition. I then performed a final health check to confirm the ingestion pipeline, alerts, and dashboard were still working."

10. Accuracy & Scope Statement

This is a personal controlled home-lab project completed by Norbert Endoh. It demonstrates hands-on practice with SOC workflows at lab scale and is not presented as production SOC employment or enterprise-scale operational experience. All controlled security activity used for validation was generated within the lab environment.

PROJECT STATUS: COMPLETE